Assessment of open surgery suturing skills via markerless hand tracking: a multistage approach using 3D hand skeleton and kinematic metrics.

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Journal: International journal of computer assisted radiology and surgery | **Year:** 2026 | **DOI:** 10.1007/s11548-026-03770-5

Link: https://doi.org/10.1007/s11548-026-03770-5 (open with institutional access)

Abstract

The transition toward competency-based medical education requires scalable, objective surgical skill assessment. While sensor-based approaches remain burdensome, video-based deep learning models often lack the interpretability required for formative feedback. We introduce a fully automated, markerless framework for assessing vascular open surgery suturing skills providing meaningful interpretation. Our pipeline leverages 3D hand tracking coupled to an LSTM-attention network for precise temporal segmentation. By merging explicit kinematic metrics with latent deep features, we develop a voting ensemble classifier categorizing surgeons into three skill levels. We further evaluated the impact of ground-truth reliability by comparing models trained on single-assessor versus multi-assessor consensus labels. The framework achieved 96.4% segmentation accuracy and 72.2 ± 0.5% accuracy (F1 = 0.73) under leave-one-video-out cross-validation (LOVCV). To go deeper in assessing cross-surgeon generalization, we additionally report a strict leave-one-subject-out (LOSO) protocol, under which accuracy decreases to 52.3%. This gap quantifies the surgeon-specific component captured by video-level splits and reframes the system as a tool for longitudinal, per-trainee skill tracking rather than zero-shot scoring of unseen surgeons. Phase-stratified analysis uncovers that experts exhibit significantly higher jerk and velocity, reflecting decisive ballistic motor planning. Training on consensus labels yields a 15-20% performance increase over single-assessor baselines. Using a single camera and no physical markers, this framework offers a cost-effective alternative to hardware-heavy simulators, enabling democratized, interpretable feedback.

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